

Harikumar N

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Education

National Institute of Technology, Rourkela, Integrated MSc Physics 2022 – 2027

CGPA: 6.99/10

Kendriya Vidyalaya , Pattom, Higher Secondary 2021

- Percentage: 96.8
- **Academic Excellence Award:** Received academic excellence award from KVS Sangathan for being in top 1.5 percent of the students who wrote CBSE 2021 BOARD Exam

Experience

National Centre for Radio Astronomy, Master Thesis Intern May 2026 – Present

- Studying Type-I quasar accretion disk variability using ZTF time-series photometry and SDSS survey data.

Leiden University, Netherlands, Remote Research Intern May 2025 – Dec 2025

- Developed a structured catalog of absolute spectral-flux calibrators for VLTI/MATISSE and future infrared instruments, enabling precise calibration for high-resolution interferometric observations.
- Conducted comparative analysis of existing Cohen et al. calibrators and new flux models using Python and Pandas. Built visualizations with Matplotlib for spectral comparisons, anomaly detection, and model validation.

Indian Institute of Technology, Hyderabad, Remote Research Intern June 2024 – Feb 2025

- Modelled and analysed 53 archival UV spectra of Seyfert galaxy ESO 141G55 from the IUE satellite. Used JAVELIN CODE to perform reverberation mapping and found its variability.
- Utilized HEASOFT software for data reduction and modeling, employing power-law, Lorentzian models to fit spectral features.
- Quantified flux for both continuum and spectral lines using the cflux model and analyzed

Publications

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- **Study of UV Line and Continuum Variabilities in the Broadline Seyfert 1 Galaxy ESO 141-G55** <https://doi.org/10.1002/asna.70093>

Astronomische Nachrichten | *Astronomy Journal* , 2026.

Leadership / Extracurricular

AstroNITR, Astrophotography Head 2024 - 2025

- Stacked and processed images of celestial objects like Orion Nebula, Moon, Jupiter, Saturn.
- Conducted a three-day outreach event for school students as part of the National Space day celebration in which more than 300 students participated. Also coordinated STELLAR NIGHT™ in INNOVISION (Tech fest of NIT ROURKELA) 2023 and 2024, a sky gazing event organised by AstroNITR.

Technical Skills

Programming Languages: Python (NumPy, Matplotlib, Pandas), SQL

Astronomical Data Analysis: *HEASOFT/XSPEC* – Used for spectral fitting and modeling of AGN spectra. Experience with power-law, Lorentzian models.

Image Processing and Astrophotography:

- **OpenCV, Matlab** – Channel splitting (RGB, HSV), histogram equalization, noise reduction, and contrast enhancement in astronomical images.
- **DeepSkyStacker (DSS)** – Stacking and preprocessing of deep-sky images to enhance faint details.
- **Registax 6** – Wavelet processing for planetary image sharpening and detail enhancement.

Software Tools:

- *VS Code, Jupyter Notebook* – IDEs for scripting
- *LaTeX* – Writing research papers, reports, and presentations.

Additional Courses / Certifications

BRICS Astronomy IDIA Data Analytics Training Program 2025

- Participated in an 8-week training program focused on data-intensive astronomy and machine learning.
- Topics include Python for data analysis, time-series methods, and ML applications for large-scale astronomical datasets.

An Observational Overview of Active Galactic Nuclei | N. Brandt 2024
(Youtube)
Playlist Link

Gained comprehensive knowledge of AGN structure, classification, and observational techniques. Studied key topics such as AGN unification models, broad and narrow line regions, relativistic jets, and cosmic evolution. Developed insights into multiwavelength surveys, AGN feedback mechanisms, and black hole accretion processes.

Astronomy: Exploring Time and Space, | Coursera 2024

- Completed the course Astronomy: Exploring Time and Space by the University of Arizona, gaining insights into astronomical phenomena, the evolution of the universe, and modern observational techniques.

Relevant Reading

Active Galactic Nuclei by Bradley Peterson

Acquired a thorough understanding of AGN physics, including accretion processes, emission mechanisms, and the unified model. Learned about observational techniques across the electromagnetic spectrum and the role of AGNs in galaxy evolution.